

Name:.....

Student
Number

Teacher:.....



Pymble Ladies' College

Mathematics Extension 2

HSC Trial Examination

Term 3 2024

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning or calculations

Total marks 100

Section 1 – 10 marks (pages 1-4)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5-12)

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section
- Answer each question in the appropriate space in the Answer Booklet. Extra writing pages are included at the end of each question.

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Section I

10 marks

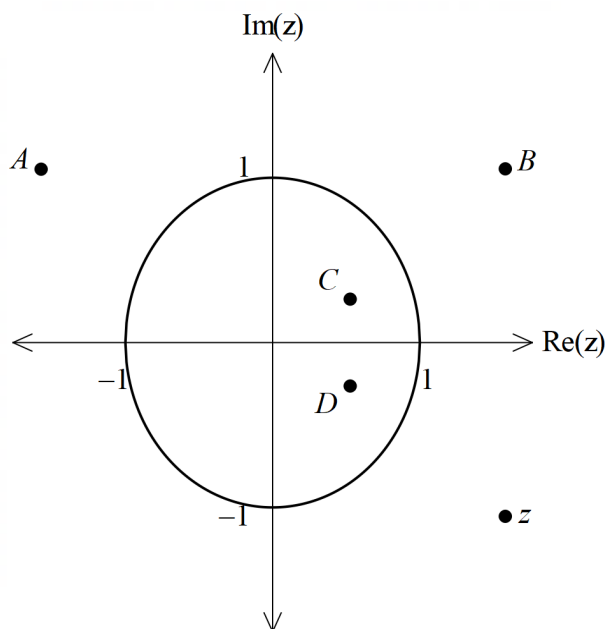
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. The velocity of a particle moving in simple harmonic motion along the x -axis is given by $V^2 = 60 + 8x - 4x^2$.
What is the centre of motion?
- (A) $x = -5$
(B) $x = 3$
(C) $x = 1$
(D) $x = 60$
2. The diagram shows the complex number z in the fourth quadrant on the Argand diagram. The modulus of z is 2.

Which of the points marked A , B , C or D best shows the position of $\frac{1}{z}$?



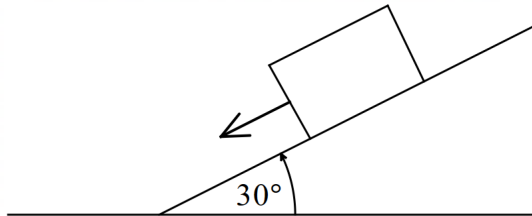
- (A) Point A
(B) Point B
(C) Point C
(D) Point D

3. Which expression is equal to $\int \tan^3 x \sec^2 x \, dx$?
- (A) $\frac{\tan^4 x}{4} + c$
- (B) $\tan^4 x + c$
- (C) $\frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c$
- (D) $\tan^5 x + \tan^3 x + c$
4. What value of z satisfies $z^2 = 7 + 24i$?
- (A) $4 + 3i$
- (B) $-4 + 3i$
- (C) $-3 + 4i$
- (D) $3 + 4i$
5. Consider the complex, non-real cube roots of unity w and w^2 .
What is the value of $(1 - w + w^2)(1 + w - w^2)$?
- (A) 0
- (B) 1
- (C) 2
- (D) 4
6. What is the size of the acute angle θ between the vectors $\underline{a} = 2\underline{i} - \underline{j} - \underline{k}$ and $\underline{b} = 2\underline{i} - 2\underline{k}$?
- (A) $\theta = \frac{\pi}{6}$
- (B) $\theta = \frac{\pi}{5}$
- (C) $\theta = \frac{\pi}{4}$
- (D) $\theta = \frac{\pi}{3}$

7. A particle has initial velocity $2\hat{i} \text{ ms}^{-1}$. At time t seconds it has acceleration $(2t-3)\hat{i} + \frac{\pi}{2} \cos\left(\frac{\pi}{2}t\right)\hat{j} \text{ ms}^{-2}$.

When is the particle at rest?

- (A) Never.
 (B) At time $t = 1$ second only.
 (C) At time $t = 2$ seconds only.
 (D) At time $t = 1$ and $t = 2$ seconds.
8. A 10 kg box on a plane inclined at an angle of 30° to the horizontal is undergoing uniform acceleration of 1.5 m/s^2 down the plane. Take the acceleration g due to gravity to be 9.8 m/s^2 .



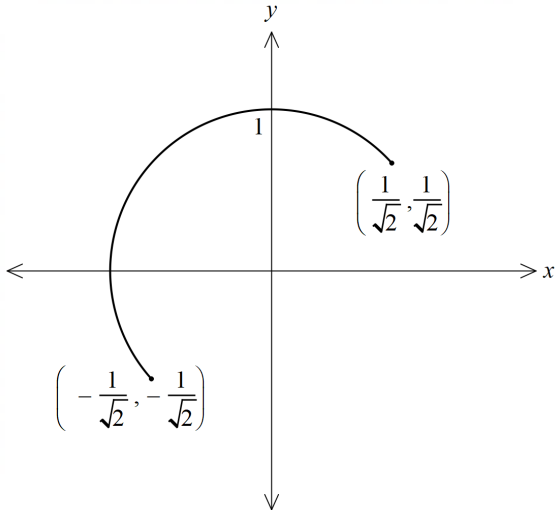
What is the magnitude of the frictional force resisting the motion of the box?

- (A) 34 N
 (B) 64 N
 (C) 70 N
 (D) 100 N
9. The points A , B and C are collinear where $\overrightarrow{OA} = \hat{i} - \hat{j}$, $\overrightarrow{OB} = -3\hat{j} - \hat{k}$ and $\overrightarrow{OC} = 2\hat{i} + a\hat{j} + b\hat{k}$ for some constants a and b . What are the values of a and b ?
- (A) $a = -1$ and $b = -1$
 (B) $a = -1$ and $b = 1$
 (C) $a = 1$ and $b = -1$
 (D) $a = 1$ and $b = 1$

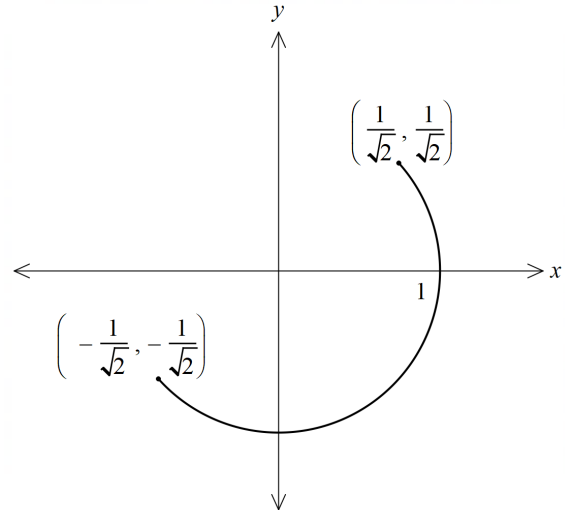
10. Which diagram best shows the curve described by the position vector

$$\underline{r}(t) = \sin(t)\underline{i} - \cos(t)\underline{j} \text{ for } \frac{\pi}{4} \leq t \leq \frac{5\pi}{4}?$$

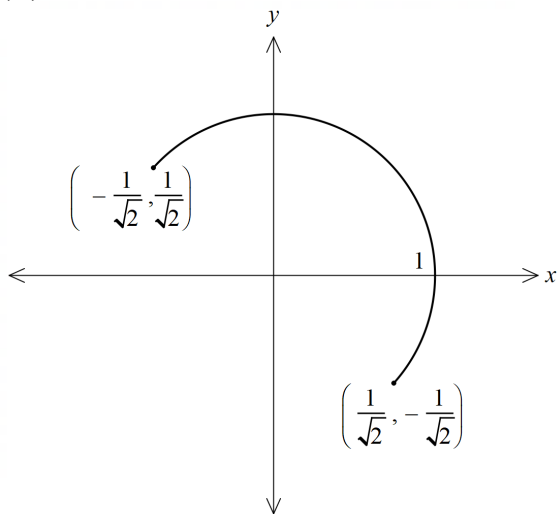
(A)



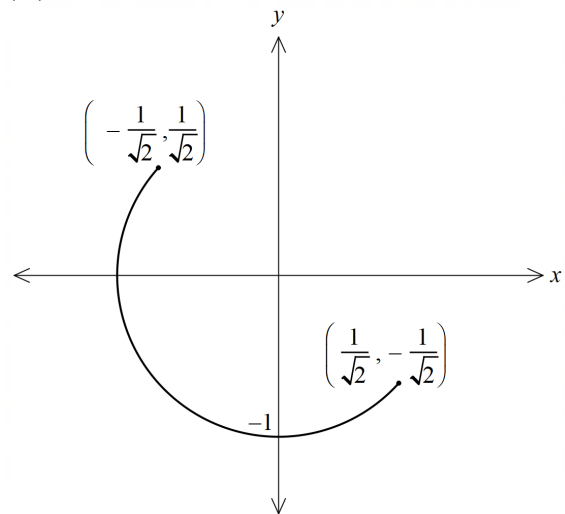
(B)



(C)



(D)



Section II

90 marks

Attempt Questions 11-16

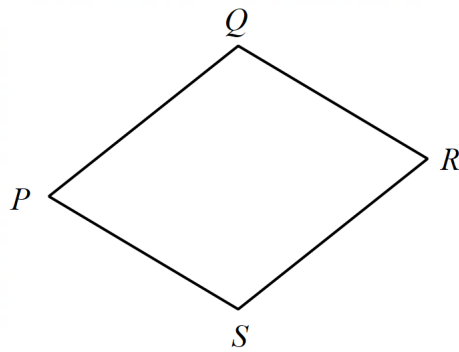
Allow about 2 hours and 45 minutes for this section.

Answer these questions in the Answer Book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (17 marks)	Marks
(a) Express $2\sqrt{2} e^{\frac{-3\pi}{4}i}$ in the form $x + iy$.	2
(b) A particle starts at rest from the origin and moves in a straight line such that its displacement x metres at time t seconds is determined by the equation $\ddot{x} = -25x$. How long will it take for the particle to first return to the origin?	2
(c) Consider the two points $A(3, 3, 3)$ and $B(3, -3, 3)$.	
(i) Find $\overrightarrow{AB}$.	1
(ii) Find $ \overrightarrow{AB} $.	1
(iii) Find $\angle AOB$. Give your answer correct to the nearest degree.	2
(d) The function $f(x)$ is defined by $f(x) = \frac{6x + 4}{(x-1)(x+1)(2x+3)}.$	
(i) Express $f(x)$ in terms of partial fractions.	2
(ii) Hence, or otherwise, find $\int \frac{3x+2}{(x-1)(x+1)(2x+3)} dx$.	2
(e) (i) Find the two square roots of $2i$, giving the answers in the form $x + iy$, where x and y are real numbers.	2
(ii) Hence, or otherwise, solve $2z^2 + 2\sqrt{2}z + 1 - i = 0$, leaving answers in the form $x + iy$.	3

End of Question 11

Question 12 (15 marks)**Marks**

(a) $PQRS$ is a parallelogram. Given that $\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\overrightarrow{QR} = 5\mathbf{i} - 2\mathbf{k}$

(i) Show that $PQRS$ is a rhombus.

2

(ii) Find the exact area of $PQRS$.

3

(b) Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{x^3}{\sqrt{1-x^2}} dx$.

4

(c) Prove by mathematical induction that $3n^2 - 3n \leq 2^n - 1$ for $n \geq 7$.

3

(d) Use the t -formula to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{3 - \cos x}$.

3**End of Question 12**

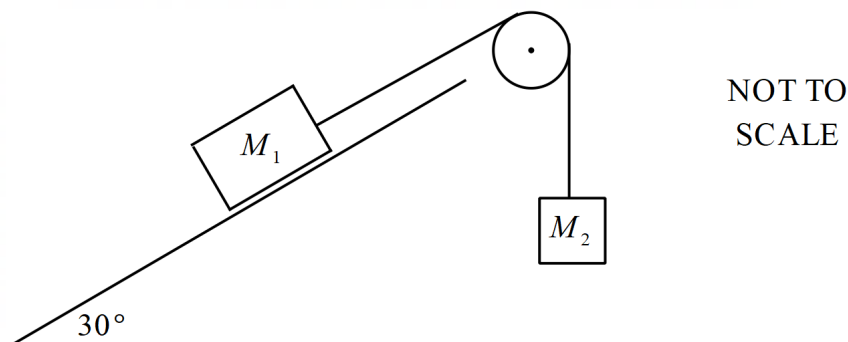
Question 13 (14 marks)**Marks**

- (a) A particle is initially 2 metres to the left of the origin, moving to the left at a speed of 3 m/s. The acceleration of the particle is given by $a = v + v^3$, where v is the velocity of the particle. **3**
Find an expression for x , the displacement of the particle, in terms of v .
- (b) Consider the statement “If Sally studies, she will pass the test.”
- (i) Write down the converse of the statement. **1**
- (ii) Write down the contrapositive of the statement. **1**
- (c) (i) The complex number z is such that $\arg(z + 2i) = \tan^{-1} \frac{1}{2}$. **2**
It is given that $z = x + iy$, where x and y are real and $x > 0$.
Find an equation for y in terms of x .
- (ii) The complex number z satisfies $-\frac{\pi}{2} \leq \arg(z + 2i) \leq \tan^{-1} \frac{1}{2}$. **2**
The region R is the locus of z . In your answer booklets, sketch the region R on the given Argand diagram.
- (iii) z_1 is the point in R at which $|z|$ is minimum. Calculate the exact value of $|z_1|$. **1**
- (iv) Express z_1 in the form $a + ib$ where a and b are real. **1**
- (d) The equations of two lines are $\underline{r} = 2\underline{i} + 3\underline{j} + 3\underline{k} + \lambda(\underline{i} - 2\underline{j} + \underline{k})$ and **3**
 $\underline{u} = \underline{i} + 11\underline{j} - 4\underline{k} + \mu(-\underline{i} + 3\underline{j} - 2\underline{k})$.
Show that the lines intersect, stating the point of intersection.

End of Question 13

Question 14 (16 marks)**Marks**

- (a) Two masses, M_1 kilograms and M_2 kilograms, are attached by a light inextensible string. The string is placed over a smooth pulley, and the mass M_1 rests on a smooth frictionless plane inclined at 30° to the horizontal as shown. The mass M_2 is suspended vertically by the string.

3

The mass M_1 accelerates down the inclined plane at 2 m/s^2 , and $g = 10 \text{ m/s}^2$.

Given $M_1 = kM_2$, find the value of k .

- (b) Evaluate $\int_0^4 \frac{1}{\sqrt{1+\sqrt{x}}} dx$.

3

Question 14 continues on page 9

- (c) A missile is fired from point O with speed V m/s at an angle α above the horizontal where $\frac{\pi}{4} < \alpha < \frac{\pi}{2}$. The missile moves in a vertical plane under gravity where the acceleration due to gravity is g m/s². At time t seconds the position vector of the missile relative to O is $\underline{r}(t) = (Vt \cos \alpha)\underline{i} + \left(Vt \sin \alpha - \frac{1}{2}gt^2\right)\underline{j}$.

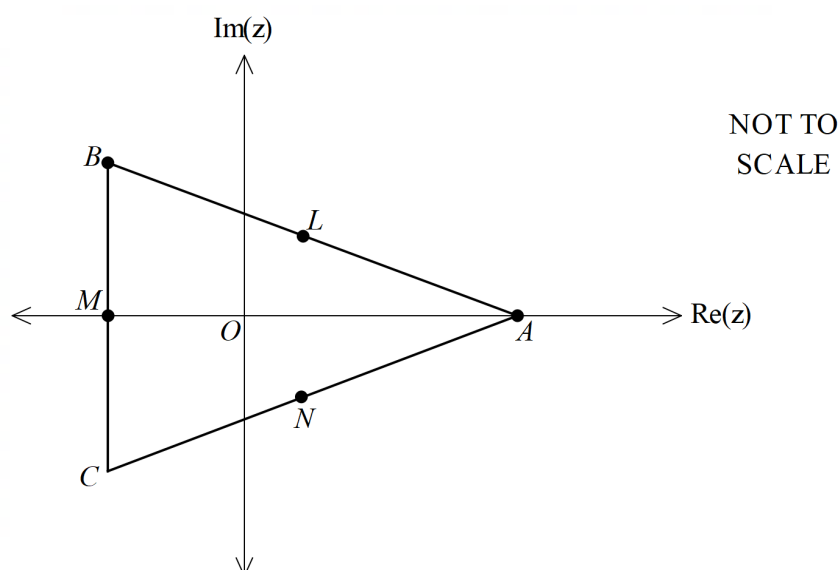
- (i) Show that the greatest height reached by the missile is $H = \frac{V^2 \sin^2 \alpha}{2g}$ metres 3
and that when the missile is at its greatest height its angle of elevation from O is $\tan^{-1}\left(\frac{1}{2}\tan \alpha\right)$.
- (ii) When the missile reaches a height of h metres its angle of inclination to the horizontal falls to $\frac{\pi}{4}$. Show that $\frac{h}{H} = 2 - \operatorname{cosec}^2 \alpha$. 3

- (d) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 1
- (ii) Hence or otherwise, to find the value of $\int_0^{\frac{\pi}{2}} \frac{\sin x}{e^{\frac{\pi}{2}} + e^x} dx$. 3

End of Question 14

Question 15 (15 marks)**Marks**

- (a) The cube roots of unity are represented on the Argand diagram below by the points A , B and C .

3

The points L , M and N are the midpoints of the line segments AB , BC and CA respectively. Determine a degree 6 polynomial equation with integer coefficient whose roots are the complex numbers represented by the points A , B , C , L , M and N .

- (b) (i) Differentiate $(16+t^2)^{\frac{3}{2}}$ with respect to t .

1

- (ii) Let $I_n = \int_0^3 t^n \sqrt{16+t^2} dt$ for integers $n \geq 1$. Show that, for $n \geq 3$,
- $$(n+2)I_n = 125 \times 3^{n-1} - 16(n-1)I_{n-2}.$$

3

- (c) A particle of mass m is projected vertically upward under gravity with speed u in a medium in which the resistance is mk times the speed, where k is a positive constant.

4

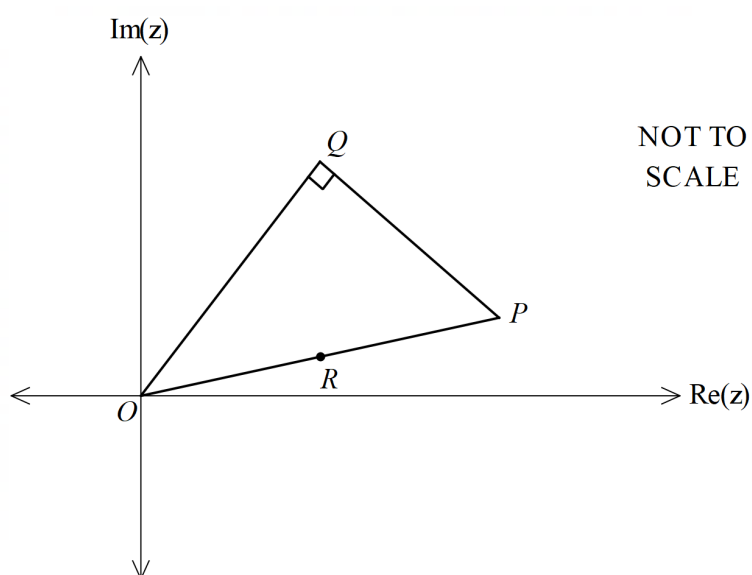
If the particle reaches its greatest height H in time T , show that $u = gT + kH$.

Question 15 continues on page 11

Question 15 continued.

Marks

- (d) In the Argand diagram below, OPQ is a triangle which is right-angled at Q . The point R is the midpoint of OP .



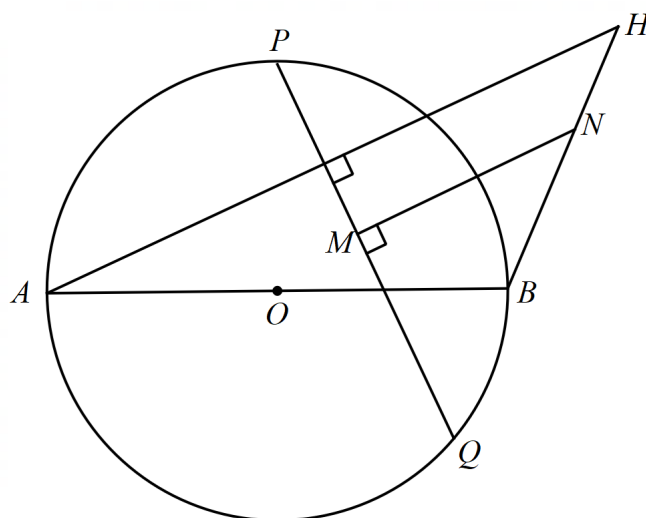
- (i) If $\overrightarrow{OP} = z$ and $\overrightarrow{OQ} = w$ show that $\overrightarrow{OR} = \frac{1}{2}(1 - ki)w$ where k is a constant, $k > 0$. **2**
- (ii) Express $\overrightarrow{RQ}$ in terms of w and hence show $|\overrightarrow{OR}| = |\overrightarrow{RQ}|$. **2**

End of Question 15

Question 16 (13 marks)

Marks

(a)



NOT TO
SCALE

In the diagram, AB is a diameter of a circle with centre O and PQ is a chord of the circle that is not perpendicular to AB . The perpendicular from A to PQ is produced to the point H outside the circle. M is the midpoint of PQ and N is the point on BH such that $MN \perp PQ$. Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OP} = \underline{p}$, $\overrightarrow{OQ} = \underline{q}$ and $\overrightarrow{AH} = \underline{h}$.

- (i) By writing $\overrightarrow{OM}$ in terms of $\underline{p}$ and $\underline{q}$, show that the points O , M and N are collinear. 2
- (ii) If $\overrightarrow{BN} = \lambda \overrightarrow{BH}$ for some scalar λ , express $\overrightarrow{ON}$ in terms of $\underline{a}$, $\underline{h}$ and λ . 2
- (iii) Hence, show that N is the midpoint of BH . 3

- (b) (i) By considering the graph of $y = \frac{1}{x\sqrt{x}}$, or otherwise, show that for all positive integers $k \geq 1$, $\frac{1}{(k+1)\sqrt{k+1}} < \frac{2}{\sqrt{k}} - \frac{2}{\sqrt{k+1}}$. 3
- (ii) Hence, use Mathematical Induction to show that for all positive integers $n \geq 2$, $\frac{1}{1\sqrt{1}} + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \dots + \frac{1}{n\sqrt{n}} < 3 - \frac{2}{\sqrt{n}}$. 3

END OF PAPER



Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

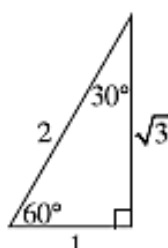
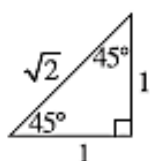
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

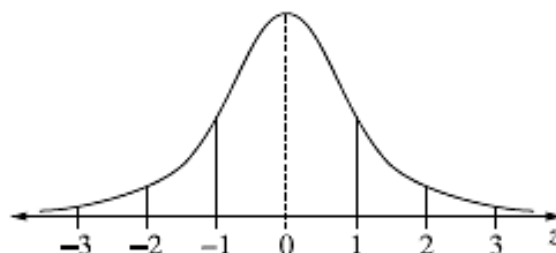
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^n C_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Name: *Solutions*

Student
Number

Teacher:.....



Pymble Ladies' College

Mathematics Extension 2

HSC Trial Examination

Term 3 2024

ANSWER BOOKLET

- Put your name, Teacher's name and Student Number on this Booklet.
- For Questions 1-10 use the Multiple-Choice Answer Sheet provided.
- Answer Questions 11-16 in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Write using a non-erasable black or blue pen. Black is preferred.
- Extra writing space is provided.
- Calculators approved by NESA may be used.

Mathematics Extension 2 HSC Trial Examination

Name:.....

Student Number:

Teacher's Name:.....

Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

1 A ☐ B ☐ C ☒ D ☐

2 A ☐ B ☐ C ☒ D ☐

3 A ☒ B ☐ C ☐ D ☐

4 A ☒ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☒

6 A ☒ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☒ D ☐

8 A ☒ B ☐ C ☐ D ☐

9 A ☐ B ☐ C ☐ D ☒

10 A ☐ B ☐ C ☒ D ☐

90 marks

Allow about 2 hours and 45 minutes for this section.

Extra writing space is available at the end of each question.

Marks

2

2

1

(c)

$$(i) \quad \vec{AB} = 0\hat{i} - 6\hat{j} + 0\hat{k}$$

1 r/w

$$(ii) \quad |\vec{AB}| = \sqrt{(-6)^2} = 6$$

(2) 1 r/w

$$(iii) \quad \cos \angle AOB = \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OA}| \times |\vec{OB}|}$$

2

$$= \frac{9 - 9 + 9}{\sqrt{3^2 + 3^2 + 3^2} \sqrt{3^2 + 3^2 + 3^2}}$$

$$= \frac{9}{27}$$

$$= \frac{1}{3}$$

$$\cos \angle AOB = \frac{1}{3}$$

$$\angle AOB = 70^\circ 31' 43.61''$$

$$\approx 71^\circ$$

2	correct
1	one error

Add
lines

(d)

$$(i) \frac{6x+4}{(x-1)(x+1)(2x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{2x+3}$$

2

$$6x+4 = A(x+1)(2x+3) + B(x-1)(2x+3) + C(x-1)(x+1)$$

$$\text{When } x=1 \quad \text{When } x=-1 \quad x=-3/2$$

$$10 = 10A \quad -2 = -2B \quad -5 = \frac{5}{4}C$$

$$\therefore A=1 \quad \therefore B=1 \quad \therefore C=-4$$

$$\therefore \frac{6x+4}{(x-1)(x+1)(2x+3)} = \frac{1}{x-1} + \frac{1}{x+1} - \frac{4}{2x+3}$$

2	Correct
1	Two of the 3 parts correct

$$(ii) \int \frac{3x+2}{(x-1)(x+1)(2x+3)} dx = \frac{1}{2} \int \left(\frac{1}{x-1} + \frac{1}{x+1} - \frac{4}{2x+3} \right) dx$$

$$= \frac{1}{2} \ln|x-1| + \frac{1}{2} \ln|x+1| - \ln|2x+3| + C$$

2	Correct
1	one error

(e)

(i)

2

$$(a+ib)^2 = 2i$$

$$a^2 - b^2 + 2abi = 2i$$

Roots are

$$a^2 - b^2 = 0$$

 $1+i, -1-i$

$$2ab = 2$$

$$ab = 1$$

$$\therefore a = \pm 1$$

$$b = \pm 1$$

$$b = \pm 1$$

$$b = \pm 1$$

$$b = \pm 1$$

$$b = \pm 1$$

(ii)

$$z = \frac{-2\sqrt{2} \pm \sqrt{8 - 4(2)(1-i)}}{4}$$

(2) 3

$$= \frac{-2\sqrt{2} \pm \sqrt{8i}}{4}$$

$$= \frac{-2\sqrt{2} \pm \sqrt{8i}}{4}$$

$$= \frac{-2\sqrt{2} \pm 2\sqrt{2i}}{4}$$

$$= \frac{-\sqrt{2} \pm (1+i)}{2}$$

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End of Question 11

Question 12 (15 marks)

Marks

(a)

(i) $|\vec{PQ}| = \sqrt{2^2 + 3^2 + (-4)^2} = \sqrt{29}$ Since $|\vec{PQ}| = |\vec{QR}|$,
adjacent sides of a parallelogram are equal in length, so all four sides are equal in length, $\therefore PQRS$ is a rhombus.

$|\vec{QR}| = \sqrt{5^2 + (-2)^2} = \sqrt{29}$

2	correct
1	Finds one modulus

(ii) $\vec{PR} = \vec{PQ} + \vec{QR} = 7\hat{i} + 3\hat{j} - 6\hat{k}$

3

$\vec{QS} = \vec{PS} - \vec{PQ} = 3\hat{i} - 3\hat{j} + 2\hat{k}$

Area $PQRS = \frac{1}{2} \times |\vec{PR}| \times |\vec{QS}|$
 $= \frac{1}{2} \sqrt{49 + 9 + 36} \times \sqrt{9 + 9 + 4} = \sqrt{517} \text{ units}^2$

(b)

4

$\int_0^{1/\sqrt{2}} \frac{x^3}{\sqrt{1-x^2}} dx$ Let $x = \sin \theta$
 when $x = 1/\sqrt{2}$, $\theta = \pi/4$

$= \int_0^{\pi/4} \frac{\sin^3 \theta}{\sqrt{1-\sin^2 \theta}} \times \cos \theta d\theta$ $x=0$ $\theta=0$
 $dx = \cos \theta d\theta$

$= \int_0^{\pi/4} \sin^2 \theta d\theta$

$= \int_0^{\pi/4} \sin \theta (1 - \cos^2 \theta) d\theta$

$= \left[-\cos \theta + \frac{1}{3} \cos^3 \theta \right]_0^{\pi/4}$

$= -\cos \pi/4 + \frac{1}{3} \cos^3 \pi/4 - \left(-\cos 0 + \frac{1}{3} \cos^3 0 \right)$

$= -\frac{1}{\sqrt{2}} + \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 + 1 - \frac{1}{3}$

$= \frac{8 - 5\sqrt{2}}{12}$

Question 12 continues on page 8

4	Correct
3	Correct strategy one error
2	Correct substitution and some progress
1	Correct substitution

3	correct
2	one error
1	Finds diagonal lengths

(c)

3

Let the statement be $3n^2 - 3n \leq 2^n - 1$ for $n \geq 7$

when $n = 7$

LHS: $3 \times 7^2 - 3 \times 7$	RHS: $2^7 - 1$	3	Correct
$= 126$	$= 127$	2	Some progress with set up

$\therefore$ Statement is true for $n = 7$

Assume the statement true for $n = k$

i.e. $3k^2 - 3k \leq 2^k - 1 \quad k \geq 7$

RTP the statement true for $n = k + 1$

i.e. $3(k+1)^2 - 3(k+1) \leq 2^{k+1} - 1 \quad k \geq 7$

Now from assumption

$$3k^2 - 3k \leq 2^k - 1$$

$$6k^2 - 6k \leq 2^{k+1} - 2$$

$$6k^2 - 6k + 1 \leq 2^{k+1} - 1$$

$$3k^2 + 3k + 3k^2 - 9k + 1 \leq 2^{k+1} - 1$$

$$3k^2 - 9k + 1 > 6 \text{ when } k \geq 7 \text{ since } a > 0 \text{ and } k = \frac{9 \pm \sqrt{69}}{6}$$

Question 12 continues on page 9

are the roots

$$\therefore 3k(k+1) < 3k(k+1) + (3k^2 - 9k + 1) \leq 2^{k+1} - 1$$

$$\therefore 3(k+1)^2 - 3(k+1) \leq 2^{k+1} - 1$$

Hence, since the statement is true for $n = 7$, assumed true for $n = k$ and proved true for $n = k + 1$, so the statement is true for $n = 7 + 1 = 8$ and so on $\forall n$ by the principle of mathematical induction.

(d)

3

$$\int_0^{\pi/2} \frac{dx}{3 - \cos x}$$

when $x = \pi/2$ $t = \tan \pi/4 = 1$

$$= \int_0^1 \frac{1}{3 - \left(\frac{1-t^2}{1+t^2}\right)} \times \frac{1}{1+t^2} dt$$

when $x = 0$ $t = \tan 0 = 0$

$$= \int_0^1 \frac{1}{\frac{3+3t^2-1+t^2}{1+t^2}} \times \frac{1}{1+t^2} dt$$

$$= \int_0^1 \frac{dt}{2+4t^2}$$

$$= \frac{1}{2} \int_0^1 \frac{dt}{1+2t^2}$$

$$= \frac{1}{2\sqrt{2}} \left[\tan^{-1} \sqrt{2} t \right]_0^1$$

$$= \frac{1}{2\sqrt{2}} \left\{ \tan^{-1} \sqrt{2} - \tan^{-1} 0 \right\}$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1} \sqrt{2}$$

3	correct
2	one error
1	just the substitution for all parts

End of Question 12

Question 13 (14 marks)

Marks

(a)

3

when $t = 0$, $x = -2$, $v = -3 \text{ m/s}$

$$v \frac{dv}{dx} = v + v^3 \quad \therefore x = \tan^{-1} v - 2 + \tan^{-1}(3)$$

$$\frac{dv}{1+v^2} = dx$$

$$x = \int \frac{dv}{1+v^2} \quad \text{---} (*)$$

$$x = \tan^{-1} v + C$$

$$-2 = \tan^{-1}(-3) + C$$

$$C = -2 + \tan^{-1}(3)$$

3	Correct
2	One error
1	Progress to *

(b)

(i) If she passes the test, Sally studied

1

(ii) If Sally does not pass the test, then she did not study.

1

Question 13 continues on page 14

(c)

(i)

2

$$\text{Let } z = x + iy$$

$$\arg(z + 2i) = \tan^{-1}(1/2)$$

$$\arg(x + (y+2)i) = \tan^{-1} 1/2$$

$$\therefore \tan^{-1} \frac{y+2}{x} = \tan^{-1} \frac{1}{2}$$

$$\therefore \frac{y+2}{x} = \frac{1}{2}$$

$$x - 2y - 4 = 0$$

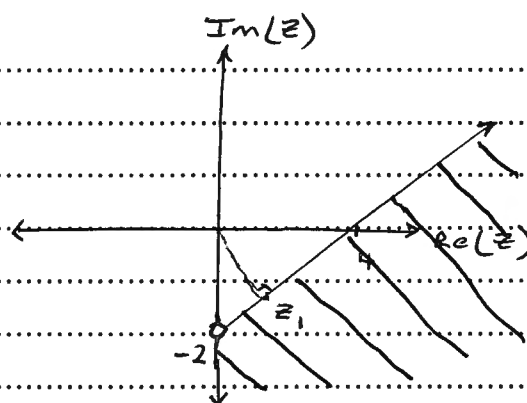
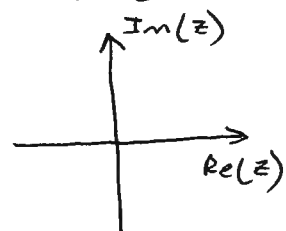
$$y = \frac{x}{2} - 2$$

2	correct
1	one error

(ii)

2

Should have



2	correct
1	one error

(iii)

1 r/w

$$|z_1| = \frac{|0 - 0 - 4|}{\sqrt{1^2 + (-2)^2}}$$

$$= \frac{4}{\sqrt{5}}$$

(iv)

1 r/w

$$z_1 = \frac{4}{5} - \frac{8}{5}i$$

$$y = \frac{x}{2} - 2$$

$$y = -2x$$

$$-2x = \frac{x}{2} - 2$$

$$x = \frac{4}{5}, y = -\frac{8}{5}$$

Question 13 continues on page 15

(d)

3

$$2 + \lambda = 1 - \mu \Rightarrow \mu + \lambda = -1 \quad \text{--- (1)}$$

$$3 - 2\lambda = 11 + 3\mu \Rightarrow 3\mu + 2\lambda = -8 \quad \text{--- (2)}$$

$$3 + \lambda = -4 - 2\mu \quad \text{--- (3)}$$

$$\textcircled{1} \times 2 - \textcircled{2} \quad -\mu = 6$$

$$\mu = -6$$

$$\text{From } \textcircled{1} \quad -6 + \lambda = -1$$

$$\lambda = 5$$

3 | Collect

2 | One error

1 | Setting up the simultaneous eq^{ns}.

$$\vec{r} = 2\vec{i} + 3\vec{j} + 3\vec{k} + 5\vec{l} - 10\vec{j} + 5\vec{k}$$

$$\vec{r} = 7\vec{i} - 7\vec{j} + 8\vec{k} \quad \therefore \text{Point of intersection}$$

$$\vec{r} = 7\vec{i} - 7\vec{j} + 8\vec{k} \quad \text{is } (7, -7, 8)$$

$\therefore$ the lines intersect.

Check when $\lambda = 5$ and $\mu = -6$.

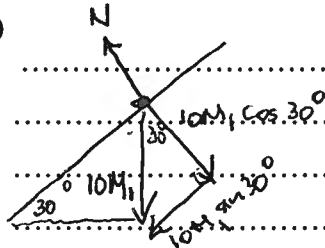
$$\begin{aligned} \textcircled{3} \Rightarrow \text{LHS} &= 3 + 5 & \text{and RHS} &= -4 - 2(-6) \\ &= 8 & &= 8 \\ & & &= \text{LHS} \end{aligned}$$

End of Question 13

Question 14 (16 marks)

Marks

(a)



$$(M_1 + M_2) \times 2 = (5M_1 - T) + (T - 10M_2)$$

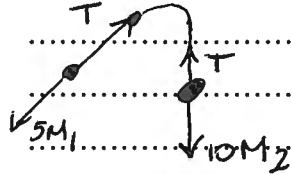
$$2M_1 + 2M_2 = 5M_1 - 10M_2$$

$$12M_2 = 3M_1$$

$$M_1 = 4M_2$$

$$\therefore k = 4$$

OR
$$\begin{cases} M_1 g \sin 30^\circ - T = 2M_1 \\ T - M_2 g = 2M_2 \end{cases}$$



3	correct
2	Equating the forces on M_1 and M_2
1	Some attempt at creating the force on M_1

(b)

$$\int_0^4 \frac{1}{\sqrt{1+\sqrt{x}}} dx$$

Let $u^2 = 1 + \sqrt{x}$

$$2u du = \frac{1}{2\sqrt{x}} dx$$

$$= \int_1^{\sqrt{3}} \frac{1}{u} \times 4u(u^2 - 1) du$$

$$4u\sqrt{x} du = dx$$

$$dx = 4u(u^2 - 1)$$

$$= 4 \int_1^{\sqrt{3}} (u^2 - 1) du$$

When $x = 4$ $u^2 = 1 + \sqrt{4}$

$$u^2 = 3$$

$$\therefore u = \sqrt{3}$$

$$= 4 \left[\frac{u^3}{3} - u \right]_1^{\sqrt{3}}$$

When $x = 0$ $u^2 = 1$

$$u = 1$$

$$= 4 \left\{ \frac{3\sqrt{3}}{3} - \sqrt{3} - \frac{1}{3} + 1 \right\}$$

$$= \frac{8}{3}$$

3	Correct
2	integrate correctly from correct substitution
1	Correct and relevant Substitution

Question 14 continues on page 20

(c)

3

$$\begin{aligned} \mathbf{r}(t) &= V \cos \alpha \, \mathbf{i} + (V \sin \alpha - \frac{1}{2} g t^2) \, \mathbf{j} \\ \mathbf{v}(t) &= V \cos \alpha \, \mathbf{i} + (V \sin \alpha - g t) \, \mathbf{j} \end{aligned}$$

$$\text{When } y = 0, \quad V \sin \alpha - g t = 0.$$

$$t = \frac{V \sin \alpha}{g}.$$

$$\begin{aligned} \text{Max height } H &= V \left(\frac{V \sin^2 \alpha}{g} \right) - \frac{1}{2} \left(\frac{V \sin \alpha}{g} \right)^2 \\ &= \frac{V^2 \sin^2 \alpha}{g} - \frac{V^2 \sin^2 \alpha}{2g} \\ &= \frac{V^2 \sin^2 \alpha}{2g}. \end{aligned}$$

$$\begin{aligned} \text{Horizontal displacement at max height } X &= V \left(\frac{V \sin \alpha}{g} \right) \cos \alpha \\ &= \frac{V^2 \sin \alpha \cos \alpha}{g}. \end{aligned}$$

$$\tan \theta = \frac{H}{X}.$$

$$= \frac{\frac{V^2 \sin^2 \alpha}{2g}}{\frac{V^2 \sin \alpha \cos \alpha}{g}}$$

$$= \frac{\sin \alpha}{2 \cos \alpha}$$

$$= \frac{1}{2} \tan \alpha.$$

$$\therefore \theta = \tan^{-1} \left(\frac{1}{2} \tan \alpha \right).$$

1: max height H .2: X & some progress on $\tan \theta$

3: correct solutions.

Question 14 continues on page 21

(c)

(ii) let β be the angle of inclination to the horizontal.

3

$$\tan \beta = \frac{y}{x} = \frac{V \sin \alpha - gt}{V \cos \alpha}$$

since $\beta = \frac{\pi}{4}$, $\frac{V \sin \alpha - gt}{V \cos \alpha} = 1$.

$$V \sin \alpha - gt = V \cos \alpha \quad (1)$$

for $y = h$, $Vt \sin \alpha - \frac{1}{2}gt^2 = h \quad (2)$

from (1), $t = \frac{V(\sin \alpha - \cos \alpha)}{g}$

(1) $\times t$ - (2) $\times 2$: $Vt \sin \alpha - 2Vt \sin \alpha = Vt \cos \alpha - 2h$

$$-Vt \sin \alpha = Vt \cos \alpha - 2h$$

$$h = \frac{Vt}{2} (\cos \alpha + \sin \alpha)$$

sub $t = \frac{V(\sin \alpha - \cos \alpha)}{g}$

$$h = \frac{V^2}{2g} (\sin^2 \alpha - \cos^2 \alpha)$$

$$H = \frac{V^2 \sin^2 \alpha}{2g}$$

$$\frac{h}{H} = \frac{\sin^2 \alpha - \cos^2 \alpha}{\sin^2 \alpha}$$

$$= 1 - \frac{\cos^2 \alpha}{\sin^2 \alpha}$$

$$= 1 - \cot^2 \alpha$$

$$= 1 - (\operatorname{cosec}^2 \alpha - 1)$$

$$= 2 - \operatorname{cosec}^2 \alpha$$

1 - apply $\tan \beta = 1$

2 - significant progress

3 - correct solutions.

Question 14 continues on page 22

(d)

(i) Let $a-x=M$ when $x=0$ $M=a$ but $x=a$ $M=0$

$$\begin{aligned} \text{RHS: } \int_0^a f(a-x) dx &= \int_a^0 f(M) (-dM) = \int_0^a f(M) dM \\ &= \int_0^a f(M) dM = \text{LHS} \end{aligned}$$

(ii)

$$\begin{aligned} I &= \int_0^{\pi} \frac{\sin x}{e^{\pi/2} + e^x} dx & 3 & \text{Correct} \\ &= \int_0^{\pi} \frac{\sin(\pi-x)}{e^{\pi/2} + e^{(\pi-x)}} dx & 2 & \text{one error} \\ I &= \int_0^{\pi} \frac{\sin x e^{-\pi/2} e^x}{e^{\pi/2} + e^x} dx & 1 & \text{Replacement of } x \text{ by } \pi-x \text{ and attempt at process} \\ e^{\pi/2} I &= \int_0^{\pi} \frac{e^x \sin x}{e^{\pi/2} + e^x} dx \\ 2e^{\pi/2} I &= \int_0^{\pi} \frac{e^{\pi/2} \sin x}{e^{\pi/2} + e^x} dx + \int_0^{\pi} \frac{e^x \sin x}{e^{\pi/2} + e^x} dx \\ &= \int_0^{\pi} \frac{(e^{\pi/2} + e^x) \sin x}{e^{\pi/2} + e^x} dx \\ &= \int_0^{\pi} \sin x dx \\ 2e^{\pi/2} I &= -[\cos x]_0^{\pi} \\ &= -(-1-1) \\ \therefore I &= e^{-\pi/2} \end{aligned}$$

End of Question 14

- (a) Vertices of ABC satisfy $z^3 - 1 = 0$
 Complex number represented by M = $-\frac{1}{2}$

3

Vertices of LMN satisfy $8z^3 + 1 = 0$

$$(z^3 - 1)(8z^3 + 1) = 0$$

$$8z^6 - 7z^3 - 1 = 0$$

OR $A(1) \quad B(-\frac{1}{2} + \frac{\sqrt{3}}{2}i) \quad C(-\frac{1}{2} - \frac{\sqrt{3}}{2}i)$
 $L(\frac{1}{4} + \frac{\sqrt{3}}{4}i) \quad N(\frac{1}{4} - \frac{\sqrt{3}}{4}i) \quad M(-\frac{1}{2})$

Quadratic satisfying BC $z^2 + z + 1 = 0$

LN $4z^2 - 2z + 1 = 0$

AM $2z^2 - z - 1 = 0$

All 6 pts

$$(z^2 + z + 1)(4z^2 - 2z + 1)(2z^2 - z - 1) = 0$$

$$8z^6 - 7z^3 - 1 = 0$$

Question 15 continues on page 26

3	Correct
2	Some progress using either method
1	Just the coordinates of the 6 pts.

(b)

$$(i) \frac{d}{dt} (16+t^2)^{3/2} = \frac{3}{2} (16+t^2)^{1/2} \times 2t$$

1 r/w

$$= 3t \sqrt{16+t^2}$$

$$(ii) I_n = \int_0^3 t^{n-1} t \sqrt{16+t^2} dt$$

3

$$* = \left[\frac{1}{3} t^{n-1} (16+t^2)^{3/2} \right]_0^3 - \frac{1}{3} \int_0^3 (n-1) t^{n-2} (16+t^2)^{3/2} dt$$

$$3I_n = \left[t^{n-1} (16+t^2)^{3/2} \right]_0^3 - (n-1) \int_0^3 t^{n-2} (16+t^2) \sqrt{16+t^2} dt$$

$$= 3^{n-1} (16+9)^{3/2} - (n-1) (16I_{n-2} + I_n)$$

$$(n+2)I_n = 125 \times 3^{n-1} - 16(n-1)I_{n-2} \text{ As req'd}$$

3	Correct
2	Significant Process
1	First working line *

Question 15 continues on page 27

(c)

$$F = ma = -mg - mkv$$

$$a = -g - kv$$

$$\frac{dv}{dt} = -g - kv$$

1 - progress to T or H.

$$\frac{dv}{g+kv} = -dt$$

2 - progress to both T & H

$$\int_u^0 \frac{dv}{g+kv} = \int_0^T -dt$$

3 - get to T & H

$$\frac{1}{k} [\ln(g+kv)]_u^0 = -T$$

4 - Constant solutions.

$$\frac{1}{k} (\ln(g+ku) - \ln(g)) = T$$

$$T = \frac{1}{k} \ln \left(\frac{g+ku}{g} \right)$$

$$\int_u^0 v \frac{dv}{g+kv} = \int_0^H -dx$$

$$LHS = \int_u^0 \frac{\frac{1}{2}(g+kv-g)}{g+kv} dv$$

$$RHS = -H$$

$$= \int_u^0 \left(\frac{\frac{1}{2}(g+kv)}{g+kv} - \frac{g}{2} \frac{1}{g+kv} \right) dv$$

$$= \frac{1}{2} \left[v - \frac{g}{k} \ln(g+kv) \right]_u^0$$

$$= \frac{1}{2} \left[\left(0 - \frac{g}{k} \ln(g) \right) - \left(u - \frac{g}{k} \ln(g+ku) \right) \right]$$

$$= \frac{1}{2} \left(\frac{g}{k} \ln \left(\frac{g+ku}{g} \right) - u \right)$$

$$\therefore H = \frac{1}{2} \left(\frac{g}{k} \ln \left(\frac{g+ku}{g} \right) - u \right)$$

$$= -\frac{1}{2} (gT - u)$$

$$kH = -gT + u \quad \therefore u = gT + kH$$

Question 15 continues on page 30

(d)

(i) $\vec{QP}$ is a 90° clockwise rotation of $\vec{OQ}$ multiplied by some scale factor
 i.e. $\vec{QP} = -ki\vec{OQ}$
 $\therefore \vec{QP} = -ki\omega$
 $\vec{OP} = \vec{OQ} + \vec{QP}$
 $= \omega - ki\omega$
 $\vec{OR} = \frac{1}{2}\vec{OP}$
 $= \frac{1}{2}(\omega - ki\omega)$
 $= \frac{1}{2}(1 - ik)\omega$

2

2	Correctly shown
1	Not fully explained

(ii) $\vec{RQ} = \vec{OQ} - \vec{OR}$

2

$= \omega - \frac{1}{2}(1 - ik)\omega$
 $= \frac{1}{2}\omega - \frac{1}{2}ik\omega$
 $\vec{OR}$ and $\vec{RQ}$ are conjugates, hence their moduli are equal

2	Correct
1	Finding $\vec{RQ}$

End of Question 15

If you use this space, clearly indicate which question you are answering.

$$t = \frac{1}{k} \ln \left(\frac{g+ku}{g+kv} \right)$$

$$e^{kt} = \frac{g+ku}{g+kv}$$

$$ge^{kt} + kve^{kt} = g+ku$$

$$v = \frac{1}{ke^{kt}} (g+ku - ge^{kt})$$

$$= \frac{(g+ku)e^{-kt} - g}{k}$$

$$\int_0^H dx = \int_0^T \frac{(g+ku)e^{-kt} - g}{k} dt$$

$$H = \left[-\frac{1}{k} (g+ku)e^{-kt} - gt \right]_0^T$$

$$= \left[-\frac{1}{k} (g+ku)e^{-kT} - gT - \left(-\frac{1}{k} (g+ku) \right) \right]$$

$$= \frac{-(g+ku)e^{-kT} + (g+ku)}{k} - \frac{gT}{k}$$

Question 16 (13 marks)

Marks

(a)

(i) $\vec{OM} = \frac{1}{2}(\vec{p} + \vec{q})$

2

$\vec{OM} \cdot \vec{PQ} = \frac{1}{2}(\vec{p} + \vec{q}) \cdot (\vec{q} - \vec{p})$

2 | correct

$= \frac{1}{2}(\vec{q} \cdot \vec{q} - \vec{p} \cdot \vec{p}) = 0$

1 | one part of proof only

since $|\vec{OP}| = |\vec{OQ}|$ (radii)

$\angle QMN + \angle OMQ = 90^\circ + 90^\circ = 180^\circ$

$\therefore O, M, N$ are collinear

(ii) $\vec{ON} = \vec{OB} + \vec{BN}$

2

$= -\vec{a} + \lambda \vec{BH}$

$= -\vec{a} + \lambda(\vec{BA} + \vec{AH})$

2 | correct

$= -\vec{a} + \lambda(2\vec{a} + \vec{h})$

1 | Some progress

$\therefore \vec{ON} = (2\lambda - 1)\vec{a} + \lambda\vec{h}$

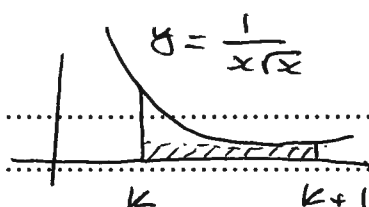
Question 16 continues on page 32

Part (a)

(iii)	$AH \perp PQ \therefore \vec{h} \cdot (\vec{q} - \vec{p}) = 0$ and O, M, N^3 are collinear $\therefore \vec{ON} \cdot \vec{PQ} = 0 \quad MN \perp PQ$	3	Correct
*	$\{(2\lambda - 1)\vec{a} + \lambda\vec{h}\} \cdot (\vec{q} - \vec{p}) = 0$	2	Significant Progress
	$(2\lambda - 1)\vec{a} \cdot (\vec{q} - \vec{p}) = 0$	1	To *
	But AB and PQ are not perpendicular $\therefore \vec{a} \cdot (\vec{q} - \vec{p}) \neq 0$ Hence $2\lambda - 1 = 0$ $\therefore \lambda = \frac{1}{2}$ and $BN = \frac{1}{2} BH$ $\therefore H$ is the midpoint of BH		

(b)

(i)



Area of shaded
rectangle = $\frac{1}{(k+1)\sqrt{k+1}}$

*	$\therefore \int_k^{k+1} \frac{1}{x\sqrt{x}} dx > \frac{1}{(k+1)\sqrt{k+1}}$	3	Correct
	$\left[-\frac{2}{\sqrt{x}} \right]_k^{k+1} > \frac{1}{(k+1)\sqrt{k+1}}$	2	Set up and method correct, or error
	$-\frac{2}{\sqrt{k+1}} + \frac{2}{\sqrt{k}} > \frac{1}{(k+1)\sqrt{k+1}}$	1	To *
	$\therefore \frac{1}{(k+1)\sqrt{k+1}} < \frac{2}{\sqrt{k}} - \frac{2}{\sqrt{k+1}}$		

Part (b)

(ii) Let the statement be $\frac{1}{\sqrt{1}} + \frac{1}{2\sqrt{2}} + \dots + \frac{1}{n\sqrt{n}} < 3 - \frac{2}{\sqrt{n}} \quad n \geq 2$

When $n = 2$.

$$\begin{aligned} \text{LHS: } \frac{1}{\sqrt{1}} + \frac{1}{2\sqrt{2}} \\ = \frac{2\sqrt{2} + 1}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{RHS: } 3 - \frac{2}{\sqrt{2}} \\ = \frac{3\sqrt{2} - 2}{\sqrt{2}} \\ = \frac{6\sqrt{2} - 4}{2\sqrt{2}} \end{aligned}$$

$\therefore \text{LHS} < \text{RHS}$

3	Correct
2	Correct set up and some progress
1	Just $n = 2$ and set u

Assume the statement is true for $n = k$

$$\text{i.e. } \frac{1}{\sqrt{1}} + \frac{1}{2\sqrt{2}} + \dots + \frac{1}{k\sqrt{k}} < 3 - \frac{2}{\sqrt{k}} \quad k \geq 2$$

RTP the statement is true for $n = k + 1$

$$\text{i.e. } \frac{1}{\sqrt{1}} + \frac{1}{2\sqrt{2}} + \dots + \frac{1}{k\sqrt{k}} + \frac{1}{(k+1)\sqrt{k+1}} < 3 - \frac{2}{\sqrt{k+1}}$$

Now

$$\begin{aligned} \frac{1}{\sqrt{1}} + \frac{1}{2\sqrt{2}} + \dots + \frac{1}{k\sqrt{k}} + \frac{1}{(k+1)\sqrt{k+1}} < 3 - \frac{2}{\sqrt{k}} + \frac{1}{(k+1)\sqrt{k+1}} \\ < 3 - \frac{2}{\sqrt{k}} + \frac{2}{\sqrt{k}} - \frac{2}{\sqrt{k+1}} \end{aligned}$$

from 1)

$$< 3 - \frac{2}{\sqrt{k+1}} \quad \text{As req'd}$$

Hence since the statement is true for $n = 2$

it is true for $n = 2 + 1 = 3$ and so on for all

$n \geq 2$ by principle of mathematical induction

END OF PAPER